Wonjae Kim

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Education

University of Maryland College Park — College Park, MD
Bachelor of Science in Computer Science, Minor in Business — *May 2026*

- **Honors & Awards:** Dean's List (2023–2024, 2024–2025), Bitcamp 2026 Razzle Dazzle Winner

Technical Skills

Languages: Python, Java, C/C++, TypeScript, JavaScript, Kotlin, Dart, OCaml
Frameworks & Libraries: Django, Celery, FastAPI, React, Flutter, TensorFlow
Databases & Infrastructure: PostgreSQL, Redis, AWS, DigitalOcean, Firebase

Experience

Dentalmon — June 2025 – Present
Founding Engineer — *College Park, MD*

- Deployed a dental insurance automation platform at a pilot dental clinic, processing over **1200 claims across 700 EOBs over two months** while reducing manual insurance posting time from **15 to 3 minutes per claim**
- Architected a distributed **OCR/LLM pipeline** with **20 payer-specific processors**, using serverless compute, Celery, and Redis to support idempotent callbacks and durable job recovery
- Secured multi-tenant patient data with **PostgreSQL row-level security**, per-user envelope encryption, and append-only access audits; enabled encrypted-field search using **HMAC blind indexes**
- Reduced median search-stage latency by **86%** (36 to 5 ms) using **PostgreSQL GIN indexes** for encrypted document lookup, measured on a 2,000-document synthetic benchmark

InZone — May 2025 – Aug 2025
Software Engineer Intern — *College Park, MD*

- Built an AI pipeline integrating ChatGPT, Meshy AI, and ElevenLabs APIs to generate **personalized 3D avatars with synthesized voices**, increasing average session time by **2x**
- Built a **Ready Player Me avatar customization flow** in a **React WebView**, supporting anonymous authentication, template selection, real-time asset equipping, and GLB rendering
- Implemented a **Firebase-backed coin economy** for voice interactions across iOS and Android, with balance validation, **atomic Firestore deductions**, conversation logging, and platform-specific in-app purchase tiers

NEMO — Feb 2025 – May 2025
Research Assistant — *College Park, MD*

- Optimized **parallel C/C++ algorithms** in stellar dynamics simulations to reduce runtime
- Ported core simulation modules to **Windows and macOS** by removing platform-specific dependencies

Hancom — Jun 2024 – Aug 2024
Software Engineer Intern — *South Korea*

- Built the **Flutter codebase adopted by the team** for Hancom Office Viewer's iOS and Android migration
- Implemented search, favorites, persistence, and responsive layouts, delivering assigned features **15% ahead of schedule**

Projects

August – Automated Course Scheduling | *Co-Founder & Lead Software Engineer* — Jun 2023 – May 2025

- Launched a Flutter/Django REST Framework scheduling app used by **220+ students**, using a **backtracking algorithm** to generate conflict-free schedules in **under 1 minute** based on course availability and student preferences
- Shipped schedule editing, customizable schedule colors, and a hangout feature for coordinating with friends, iterating on **feedback from student users**
- Built a **multithreaded Python scraper** using BeautifulSoup to synchronize UMD course offerings, seat availability, instructors, and meeting times with the Django backend

Ghostdiedie – Motion-Controlled 3D Fighter | *Team Lead & Full-Stack Developer* — Apr 2026

- Led a **4-person team** to build a webcam-controlled, two-player 3D fighting game, personally developing both the **React/Three.js frontend and FastAPI backend**
- Implemented **MediaPipe gesture controls** with player calibration, coordinate correction, visibility filtering, and cooldown tuning; built animated combat and custom face textures attached to avatar skeletons
- Synchronized player movement, **server-calculated damage**, and round countdowns via **WebSockets**; corrected player assignment and prevented premature attacks